Tuesday, November 11, 2025 2:37 AM

TEMPERATURE TRENDS OVERTIME

Step 1 . Firstly we load a data sets into excel

State	City	Temperatu	Date	Time	Year	Mont
Florida	Miami	29.97	2012-11-04	11:18	2012	Novembe
Florida	Orlando	21.62	2019-12-31	11:08	2019	December
Illinois	Peoria	35.48	2015-12-22	12:05	2015	December
California	San Franci	39.23	2015-12-16	06:43	2015	December
Texas	San Antoni	17.66	2022-08-13	18:33	2022	August
Illinois	Naperville	19.66	2018-09-12	18:25	2018	Septembe
California	San Franci	39.79	2013-10-06	06:45	2013	October
Florida	Orlando	30.67	2016-04-24	21:37	2016	April
California	Los Angele	21.31	2019-03-05	17:59	2019	March
Texas	Austin	32.43	2012-08-05	16:26	2012	August
Florida	Miami	20.37	2014-08-07	11:06	2014	August
California	San Diego	19.34	2014-03-30	19:16	2014	March
Illinois	Chicago	24.42	2012-07-19	04:55	2012	July
Florida	Tampa	8.87	2011-01-12	17:51	2011	January
Illinois	Springfield	5.75	2014-02-05	15:48	2014	February
New York	New York	23.38	2011-02-10	23:37	2011	February
Florida	Miami	14.87	2012-12-10	23:07	2012	December
Florida	Orlando	26.14	2022-06-07	22:24	2022	June
Florida	Tampa	18.92	2017-03-20	13:41	2017	March
California	Sacrament	9.88	2024-12-21	03:49	2024	December
California	Los Angele	40.66	2017-04-27	05:46	2017	April
California	San Diego	21.74	2010-01-14	09:37	2010	January
Illinois	Peoria	29.68	2023-02-14	09:15	2023	February
Illinois	Peoria	14.75	2011-03-19	11:01	2011	March
Florida	Jacksonvill	15.56	2022-02-12	19:07	2022	February
California	Los Angele	27.11	2014-03-07	06:30	2014	March

Step 2. After loading datasets then we are using conditional formatting

1)Greater Than -> Temperature
2)Less Than -> Year
3)Equal To -> Year
4)Text Contains -> State/Month/City

	State	City	Temperatu	Date	Time	Year	Mont
2	Florida	Miami	29.97	2012-11-04	11:18	2012	Novembe
3	Florida	Orlando	21.62	2019-12-31	11:08	2019	December
4	Illinois	Peoria	35.48	2015-12-22	12:05	2015	December
5	California	San Franci	39.23	2015-12-16	06:43	2015	December
6	Texas	San Antoni	17.66	2022-08-13	18:33	2022	August
7	Illinois	Naperville	19.66	2018-09-12	18:25	2018	Septembe

8	California	San Franci	39.79	2013-10-06	06:45	2013	October
9	Florida	Orlando	30.67	2016-04-24	21:37	2016	April
10	California	Los Angele	21.31	2019-03-05	17:59	2019	March
11	Texas	Austin	32.43	2012-08-05	16:26	2012	August
12	Florida	Miami	20.37	2014-08-07	11:06	2014	August
13	California	San Diego	19.34	2014-03-30	19:16	2014	March
14	Illinois	Chicago	24.42	2012-07-19	04:55	2012	July
15	Florida	Tampa	8.87	2011-01-12	17:51	2011	January
16	Illinois	Springfield	5.75	2014-02-05	15:48	2014	February
17	New York	New York	23.38	2011-02-10	23:37	2011	February
18	Florida	Miami	14.87	2012-12-10	23:07	2012	December
19	Florida	Orlando	26.14	2022-06-07	22:24	2022	June
20	Florida	Tampa	18.92	2017-03-20	13:41	2017	March
21	California	Sacrament	9.88	2024-12-21	03:49	2024	December
22	California	Los Angele	40.66	2017-04-27	05:46	2017	April
23	California	San Diego	21.74	2010-01-14	09:37	2010	January
24	Illinois	Peoria	29.68	2023-02-14	09:15	2023	February
25	Illinois	Peoria	14.75	2011-03-19	11:01	2011	March
26	Florida	Jacksonvill	15.56	2022-02-12	19:07	2022	February
27	California	Los Angele	27.11	2014-03-07	06:30	2014	March
28	Illinois	Peoria	17.49	2020-03-26	19:22	2020	March

Step 3 . And we find Average of

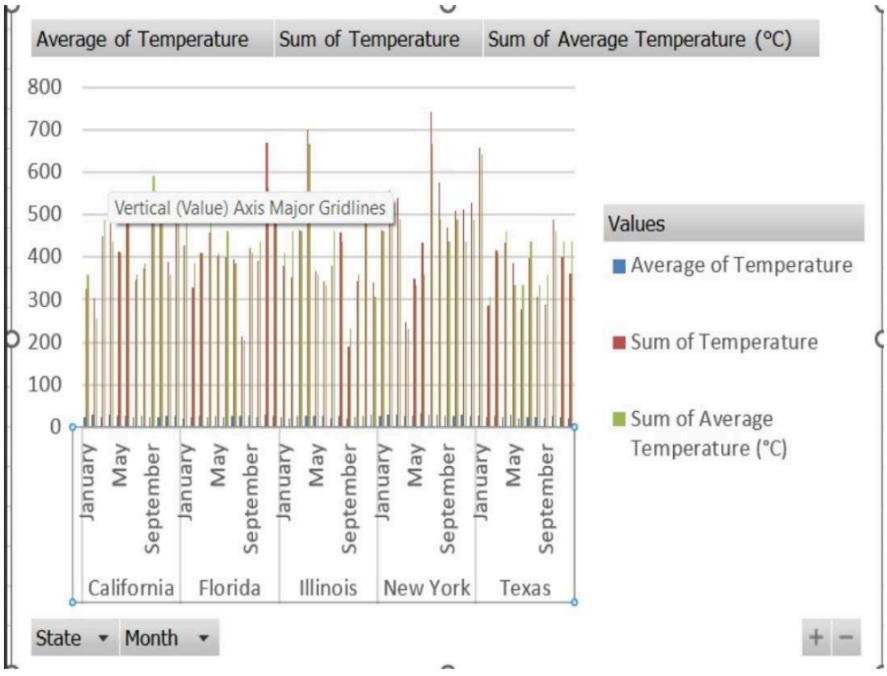
1)Temperature
2)Date

	State	City	Temperatu	Date	Time	Year	Mont	Average Temperature (°C
2	Florida	Miami	29.97	2012-11-04	11:18	2012	November	25.67926
3	Florida	Orlando	21.62	2019-12-31	11:08	2019	December	25.67926
4	Illinois	Peoria	35.48	2015-12-22	12:05	2015	December	25.67926
5	California	San Franci	39.23	2015-12-16	06:43	2015	December	25.67926
6	Texas	San Antoni	17.66	2022-08-13	18:33	2022	August	25.67926
7	Illinois	Naperville	19.66	2018-09-12	18:25	2018	Septembe	25.67926
8	California	San Franci	39.79	2013-10-06	06:45	2013	October	25.67926
9	Florida	Orlando	30.67	2016-04-24	21:37	2016	April	25.67926
10	California	Los Angele	21.31	2019-03-05	17:59	2019	March	25.67926
11	Texas	Austin	32.43	2012-08-05	16:26	2012	August	25.67926
12	Florida	Miami	20.37	2014-08-07	11:06	2014	August	25.67926
13	California	San Diego	19.34	2014-03-30	19:16	2014	March	25.67926
14	Illinois	Chicago	24.42	2012-07-19	04:55	2012	July	25.67926
15	Florida	Tampa	8.87	2011-01-12	17:51	2011	January	25.67926
16	Illinois	Springfield	5.75	2014-02-05	15:48	2014	February	25.67926
17	New York	New York	23.38	2011-02-10	23:37	2011	February	25.67926
18	Florida	Miami	14.87	2012-12-10	23:07	2012	December	25.67926
19	Florida	Orlando	26.14	2022-06-07	22:24	2022	June	25.67926
20	Florida	Tampa	18.92	2017-03-20	13:41	2017	March	25.67926
21	California	Sacrament	9.88	2024-12-21	03:49	2024	December	25.67926
22	California	Los Angele	40.66	2017-04-27	05:46	2017	April	25.67926
23	California	San Diego	21.74	2010-01-14	09:37	2010	January	25.67926
24	Illinois	Peoria	29.68	2023-02-14	09:15	2023	February	25.67926
25	Illinois	Peoria	14.75	2011-03-19	11:01	2011	March	25.67926
26	Florida	Jacksonvill	15.56	2022-02-12	19:07	2022	February	25.67926
27	California	Los Angele	27.11	2014-03-07	06:30	2014	March	25.67926
28	Illinois	Peoria	17.49	2020-03-26	19:22	2020	March	25.67926

Step 4) After completing these type of formatting then we create pivot table and charts of this Data sets

3	Row Labels	Average of Temper	Sum of Temper	Sum of Average Temperat
4	Californi	25.54320197	5185.27	5212.88978
5	January	23.30928571	326.33	359.50964
6	February	30.506	305.06	256.7926
7	March	23.67789474	449.88	487.90594
8	April	29.04235294	493.72	436.54742
9	May	25.8025	412.84	410.86816

11	July	24.74071429	346.37	359.50964
12	August	24.978	374.67	385.1889
13	September	24.16217391	555.73	590.62298
14	October	23.38238095	491.03	539.26446
15	November	27.77714286	388.88	359.50964
16	December	25.40571429	533.52	539.26446
17	**Florida**	**25.11651741**	**5048.42**	**5161.53126**
18	January	21.4725	429.45	513.5852
19	February	21.98266667	329.74	385.1889
20	March	25.564375	409.03	410.86816
21	April	24.18315789	459.48	487.90594
22	May	25.17125	402.74	410.86816
23	June	22.19722222	399.55	462.22668
24	July	26.338	395.07	385.1889
25	August	26.64625	213.17	205.43408
26	September	26.341875	421.47	410.86816
27	October	23.11117647	392.89	436.54742
28	November	30.47863636	670.53	564.94372
29	December	27.64736842	525.3	487.90594
30	**Illinois**	**24.80463918**	**4812.1**	**4981.77644**
31	January	23.7475	379.96	410.86816
32	February	19.52944444	351.53	462.22668
33	March	25.85555556	465.4	462.22668
34	April	26.87384615	698.72	667.66076
35	May	26.17	366.38	359.50964
36	June	26.39769231	343.17	333.83038
37	July	21.06944444	379.25	462.22668
38	August	26.94529412	458.07	436.54742
39	September	21.05111111	189.46	231.11334
40	October	24.435	342.09	359.50964
41	November	26.19105263	497.63	487.90594
42	December	28.37	340.44	308.15112
43	**New York**	**28.3938756**	**5934.32**	**5366.96534**
44	January	25.80833333	464.55	462.22668
45	February	29.38315789	558.28	487.90594
46	March	28.39052632	539.42	487.90594
47	April	27.49	247.41	231.11334
48	May	26.99846154	350.98	333.83038
49	June	31.04	434.56	359.50964
50	July	28.59576923	743.23	667.66076
51	August	30.31105263	575.91	487.90594
52	September	27.73647059	471.52	436.54742
53	October	26.80631579	509.32	487.90594
54	November	30.08294118	511.41	436.54742
55	December	27.77526316	527.73	487.90594
56	**Texas**	**24.34792746**	**4699.15**	**4956.09718**
57	January	26.2768	656.92	641.9815
58	February	23.74916667	284.99	308.15112
59	March	26.06625	417.06	410.86816
60	April	24.07222222	433.3	462.22668
61	May	29.71230769	386.26	333.83038
62	June	21.22769231	275.96	333.83038
63	July	23.41058824	397.98	436.54742
64	August	23.75	308.75	333.83038
65	September	20.58285714	288.16	359.50964
66	October	27.10611111	487.91	462.22668
67	November	23.57764706	400.82	436.54742
68	December	21.23764706	361.04	436.54742
69	**Grand Total**	**25.67926**	**25679.26**	**25679.26**



Step 5. After completing charts then we find data validation.

1	State	City	Temperature	Date	Time	Year	Month	Averag
2	Florida	Miami	29.97	2012-11-04	11:18	2012	November	
3	Florida	Orlando	21.62	2019-12-31	11:08	2019	December	
4	Illinois	Peoria	35.48	2015-12-22	12:05	2015	December	
5	California	San Francisco	-50	2015-12-16	06:43	2015	December	
6	Texas	San Antonio	17.66	2022-08-13	18:33	2022	August	
7	Illinois	Naperville	19.66	2018-09-12	18:25	2018	September	
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